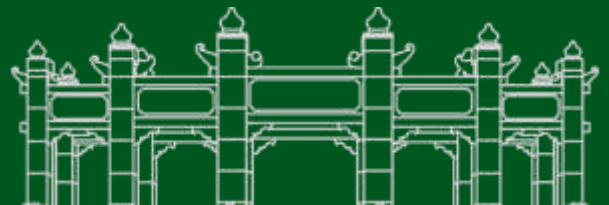
研究进展报告

丰中子Rh和Ag同位素链中的形状共存

Qin Guoyou

2026/8/28



中山大學 SUN YAT-SEN UNIVERSITY

Shape coexistence of quasiparticle states in neutron-rich Rh and Ag isotopes

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[1]*School of Physics and Astronomy, Sun Yat-sen University, Zhuhai, 519082, Guangdong, China*
[2]*Guangdong Provincial Key Laboratory of Quantum Metrology and Sensing, Sun Yat-sen University, Zhuhai, 519082, Guangdong, China*

(Dated: August 27, 2026)

We investigate low-lying one-quasiproton bandheads in odd-A $^{107-127}$Rh and $^{109-133}$Ag using configuration-constrained potential-energy-surface calculations. The blocked proton orbitals are tracked over the (β_2,γ) plane, with β_4 minimized at each mesh point. The calculated bandhead systematics follow the main trends in the available data for the $\pi 7/2^+[413]$, $\pi 1/2^-[301]$, $\pi 5/2^+[422]$, and $\pi 3/2^+[431]$ states. Strong configuration-dependent shape polarization is found in both chains. In the lighter Rh isotopes, low-lying configurations stabilize prolate, oblate, and triaxial minima. A representative case is ^{109}Rh, where the triaxial $\pi 7/2^+[413]$ ground-state configuration competes with prolate $\pi 1/2^-[301]$ and oblate $\pi 5/2^+[422]$ minima. The competition is closest in ^{115,117}Ag: the lowest $\pi 1/2^-[301]$, $\pi 7/2^+[413]$, and $\pi 3/2^+[431]$ configurations lie within about 50 keV despite their markedly different γ deformations. With increasing neutron number, the lowest positive-parity minima become nearly spherical at $N = 76$ in Rh and $N = 74$ in Ag, whereas the $\pi 1/2^-[301]$ configurations remain appreciably deformed.

I. INTRODUCTION

Neutron-rich nuclei near $A \approx 100$ undergo rapid changes in equilibrium shape [1]. In the Sr and Zr isotopes, the ground state changes from weakly deformed to strongly prolate at neutron number $N = 60$, a change described in Zr as a quantum phase transition in nuclear shape [2]. For $Z \gtrsim 41$, in Mo, Ru, and Pd, the evolution is smoother [3]. Prolate, oblate, and triaxial minima can lie close in energy and be separated by low barriers. The low-lying structure is then governed by γ softness and shape coexistence rather than by a single rigid shape. The triaxial degree of freedom also gives a pronounced contribution to the binding energy in this region. Global macroscopic–microscopic calculations find the strongest ground-state lowering from triaxiality near ^{108}Ru [4], while spectroscopic systematics suggest maximal triaxiality around $N = 68$ in ^{112}Ru and ^{114}Pd [5].

Even-even nuclei provide a clear view of these collective trends, but pairing and configuration mixing tend to obscure the underlying single-particle structure. Odd-mass neighbors probe it more directly. The unpaired nucleon occupies a specific single-particle orbital, is blocked from the pairing field, and polarizes the core toward the shape favored by that orbital [6, 7]. A fixed quasiparticle configuration can therefore stabilize one of several competing minima in a shape-soft even-even core. Odd-A nuclei may consequently exhibit low-lying one-quasiproton bandheads, and in some cases isomers, without a direct

ing triaxial and highly deformed configurations [9]. The first identification of the $1/2^-$ isomers in ^{117,119}Rh extended the systematics of the competing negative- and positive-parity states beyond $N = 70$. Triaxial relativistic Hartree–Bogoliubov calculations reproduce their nonmonotonic spacing and suggest an axial–triaxial–axial evolution, with the maximum spacing near $N \simeq 68$ [10]. In Ag, the long-lived $1/2^-$ and $7/2^+$ states are associated predominantly with the $\pi p_{1/2}$ and $\pi g_{9/2}^{-3}$ proton configurations. Precision mass measurements and laser spectroscopy established their ordering in $^{113-123}$Ag and placed the inversion at $N = 74$ or 76 [11, 12]. For ^{119}Ag, configuration-constrained PES calculations find a prolate $\pi 7/2^+[413]$ minimum at the lowest energy, with triaxial $\pi 1/2^-[301]$ and oblate $\pi 1/2^+[440]$ minima closely lying [14].

These results associate the long-lived low-lying states with proton configurations whose different spin and parity suppress direct electromagnetic decay. If the blocked configurations favor different minima, a shape change can further hinder decay by reducing the overlap of their intrinsic wave functions. This mechanism should not be conflated with conventional high-K isomerism. The quantum number K is conserved only at axial symmetry and is mixed by triaxiality. The hindrance may therefore arise from spin-parity selection and a change of quasiproton configuration, with a possible contribution from shape mismatch, rather than from K conservation alone. This differs from high-K multiquasiparticle iso-

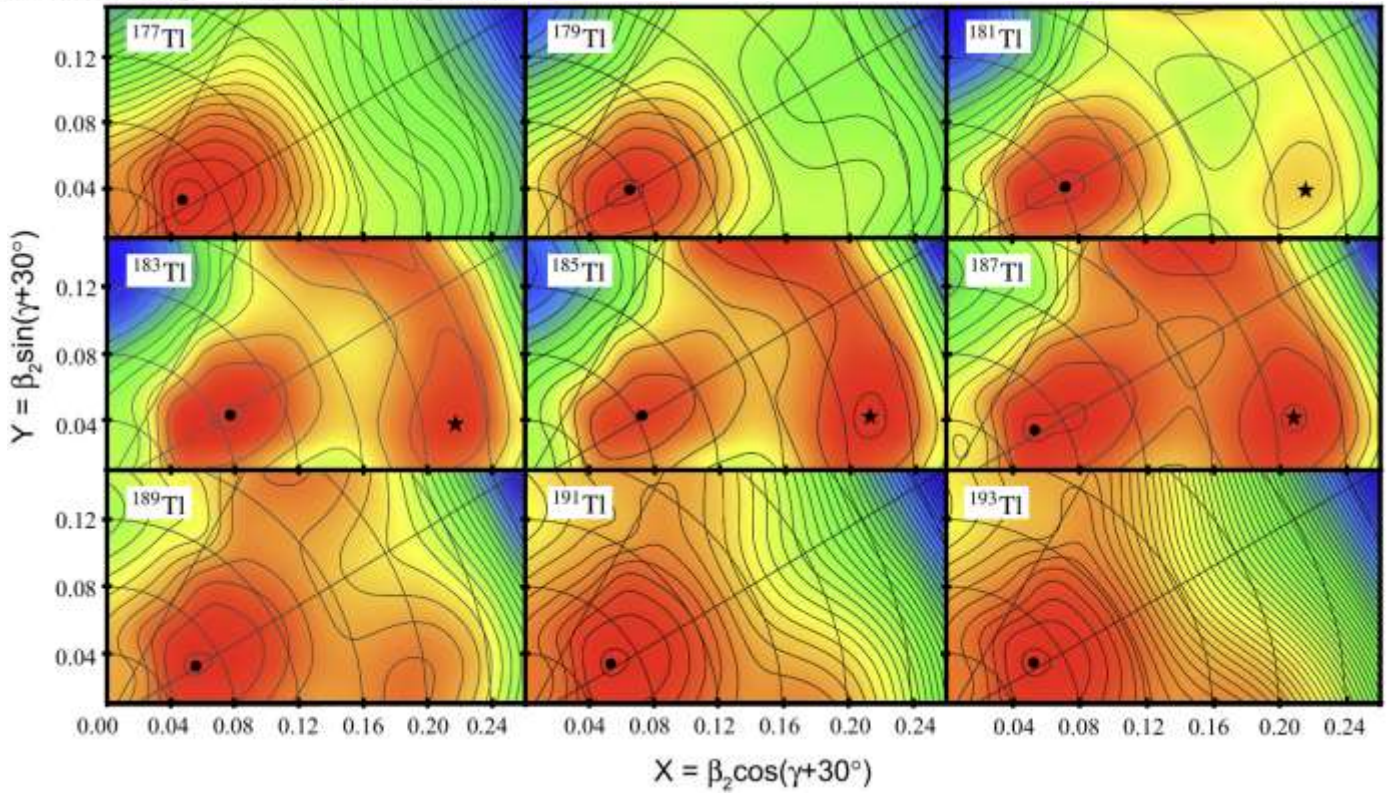
研究背景

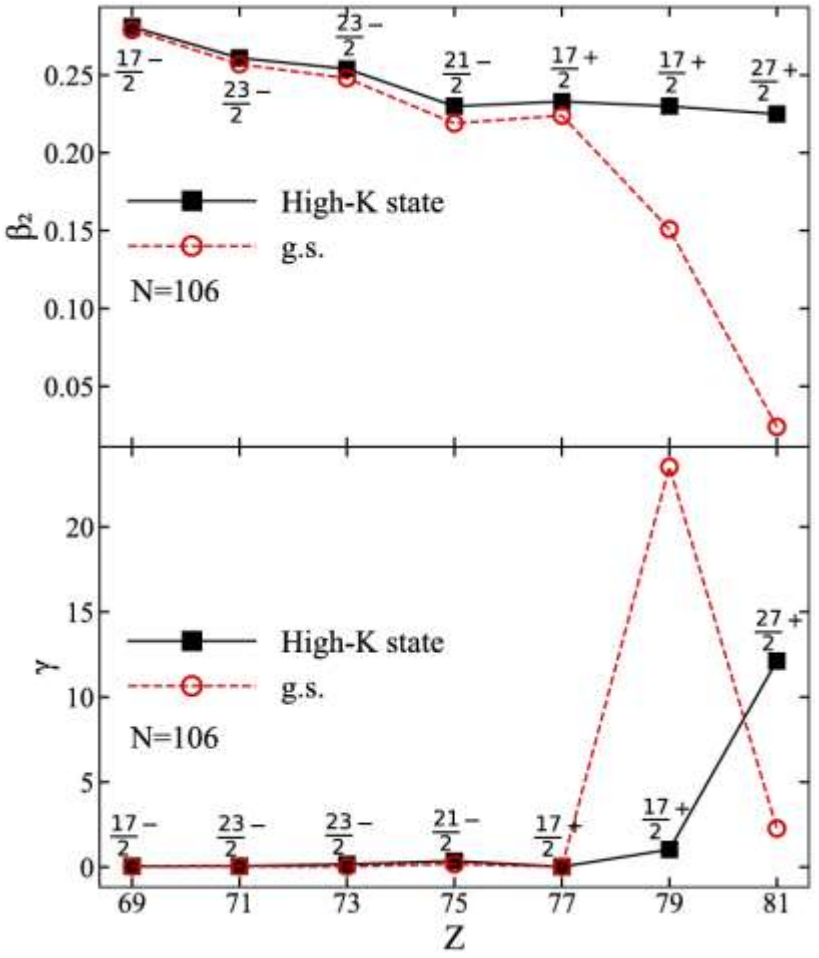
Competition between shape-coexisting $11/2^-$ states in neutron-deficient odd-A $^{181-189}$Tl

R. Y. Dong[a,b], C. F. Jiao[a,b,*], A. N. Andreyev[c,d], A. E. Barzakh[e], Z. Liu[f,g], S. Y. Zhang[h]

Shape polarization and coexistence of high-*K* three-quasiparticle states in odd-mass *N* = 106 isotones

Run-Yan Dong[1,2] · Chang-Feng Jiao[1,2]





研究背景

Binding energies of ground and isomeric states in neutron-rich ruthenium isotopes: measurements at JYFLTRAP and comparison to theory

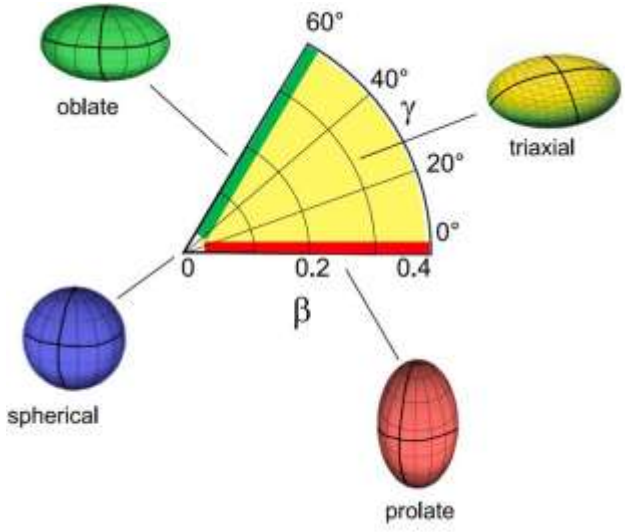
M. Hukkanen,[1,2] W. Ryssens,[3] P. Ascher,[2] M. Bender,[4] T. Eronen,[1] S. Grévy,[2] A. Kankainen,[1] M. Stryjczyk,[1]

- 报告了某大学加速器实验室使用 JYFLTRAP 双 Penning 阱质谱仪对 113, 115, 117Ru 进行的精密质量测量，并首次实现了同核异构激发能的测量 $E_x(^{113}\mathrm{Ru}^m) = 100.5(8)$ keV and $E_x(^{115}\mathrm{Ru}^m) = 129(5)$ keV.
- 117Ru 的基态采用飞行时间离子回旋共振技术测量。新的质量值比先前文献报告的值低约 36 keV，精度提高了 7 倍。利用更精确的基态质量值，双中子分离能的演化得到了进一步约束，在中子数 N = 71 之前获得了与 BSkG1 模型预测相似的趋势。

Nuclide	$T_{1/2}$ (ms)	J^π	Ref.	z	r	ME (keV)	ME$_{lit.}$ (keV)	E_x (keV)	$E_{x,lit.}$ (keV)	Diff. (keV)
^{113}Ru	800(50)	$(1/2^+)$	^{133}Cs	1	0.849 647 289(12)	−71 874.6(15)	−71 870(40)			−5(40)
^{113}Ru^m	510(30)	$(7/2^-)$	^{113}Ru	1	1.000 000 951(10)[a]	−71 774.6(18)		100.0(11)		
			^{113}Ru	2	1.000 000 963(14)[b]	−71 773.3(21)		101.3(15)		
					Final value:	−71 774.2(17)	−71 740(50)	100.5(8)[c]	131(33)	−34(50)
^{115}Ru	318(19)	$(3/2)^+$	^{133}Cs	1	0.864 742 653(23)	−66 054.7(28)	−66 105(25)			50(26)
^{115}Ru^m	76(6)	$(9/2)^-$	^{115}Ru	1	1.000 001 206(47)	−65 925.6(58)	−66 110(90)	129(5)	82(6)	184(91)
^{117}Ru	151(3)	$3/2^+$#	^{133}Cs	1	0.879 843 74(52)	−59 526(64)	−59 490(430)			−36(435)

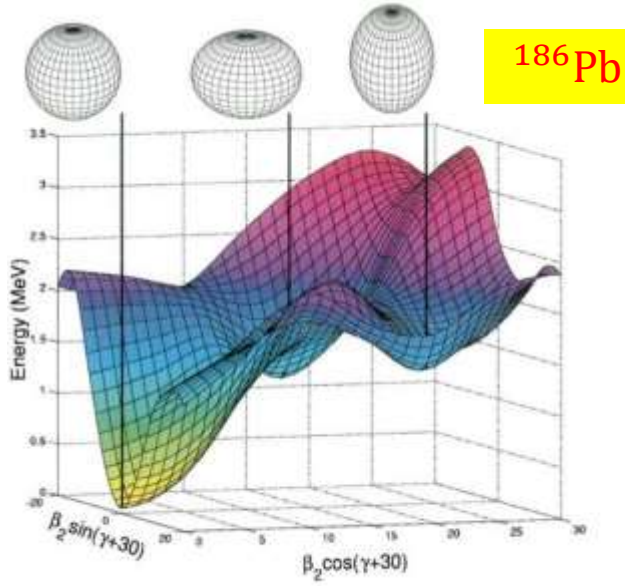
研究意义

- 锆（Z = 40）和锡（Z = 50）之间的丰中子核表现出形状共存，即激发态与不同于基态的形状相关联。
- 包含球形和轴对称的长椭、扁椭，以及没有剩余旋转对称轴的三轴形变。



S. Leoni et al., Prog. Part. Nucl. Phys. **2024**, 104119.

- 奇质子 Rh（Z = 45）和 Ag（Z = 47）同位素链位于偶偶 Ru、Pd 和 Cd 核之间，并且对这一过渡区域中依赖于组态的极化效应特别敏感。



^{186}Pb

A.N. Andreyev et al., Nature **405**, 430–433 (2000).

为什么选择Rh和Ag？

➢奇中子的 Mo（Z = 42）、Ru（Z = 44）、Pd（Z = 46）同位素链。

核素	组态	E_{cal}(keV)	E_{expt}(keV)	β_2	γ	β_4
113Pd	11/2−[505]	728		0.213	−59.971	−0.040
	3/2+[422]	527	373	0.238	−60.392	−0.025
	5/2+[422]	479	349	0.236	−60.152	−0.023
	7/2−[523]	132		0.239	−68.417	−0.028
	9/2−[514]	23	81	0.223	−59.778	−0.028
	1/2+[420]	0		0.238	−68.152	−0.029
115Pd	5/2+[402]	696		0.211	0.135	−0.021
	1/2+[431]	603		0.208	−55.044	−0.036
	9/2−[514]	432		0.218	−59.754	−0.033
	3/2+[431]	332	128	0.191	−59.561	−0.041
	1/2+[420]	295		0.243	−77.127	−0.026
	7/2−[523]	0	89	0.230	−59.917	−0.036
117Pd	1/2+[420]	650		0.229	−35.198	−0.029
	1/2+[431]	574		0.211	−59.970	−0.044
	7/2−[523]	430		0.215	−59.540	−0.038
	3/2+[431]	0		0.208	−70.006	−0.042
119Pd	3/2+[402]	438		0.099	−0.366	−0.002
	1/2+[420]	364		0.215	−87.585	−0.032
	7/2−[523]	116		0.153	−41.777	−0.030
	1/2+[431]	2		0.166	−75.149	−0.037
	3/2+[431]	0		0.170	−46.843	−0.039

组态能量随着中子数的变化剧烈改变，难以体现系统性

➢奇质子的 Nb（Z = 41）和 Tc（Z = 43）同位素链。

101Nb	7/2+[413]	956		0.340	0.772	0.003
	7/2+[413]	953		0.209	−55.070	−0.003
	5/2−[303]	377	208	0.300	−3.104	0.028
	5/2+[422]	238	0	0.305	−0.436	0.031
	3/2−[301]	0	205	0.322	−0.367	0.023
103Nb	7/2+[413]	1177		0.214	−66.426	−0.009
	7/2+[413]	820		0.347	1.807	−0.002
	5/2−[303]	406	164	0.314	5.466	0.019
	5/2+[422]	303	0	0.323	−1.747	0.024
	3/2−[301]	0	248	0.335	0.790	0.015
105Tc	1/2−[301]	973		0.257	−26.002	0.002
	3/2+[431]	885		0.216	−43.188	−0.009
	7/2+[413]	872		0.220	−79.088	−0.010
	1/2+[431]	507	322	0.290	−25.326	−0.003
	7/2+[413]	448	150	0.267	−0.037	0.026
	3/2−[301]	422		0.340	0.166	0.022
	5/2+[422]	311	85	0.343	7.198	0.014
	5/2−[303]	0	77	0.290	−21.325	0.007

与实验不够符合

组态限制的势能面(PES)

- 基于宏观-微观方法的组态限制势能面 。每个形变处的能量为

$$E(\beta_2,\gamma,\beta_4) = E_{\mathrm{LD}} + \delta E_{\mathrm{shell}} + \delta E_{\mathrm{pair}}$$

- 右边分别是液滴项、Strutinsky 壳层修正、配对修正
- 把 BCS 在单极对力下的粒子数投影扩展到 Lipkin-Nogami 近似中处理配对。

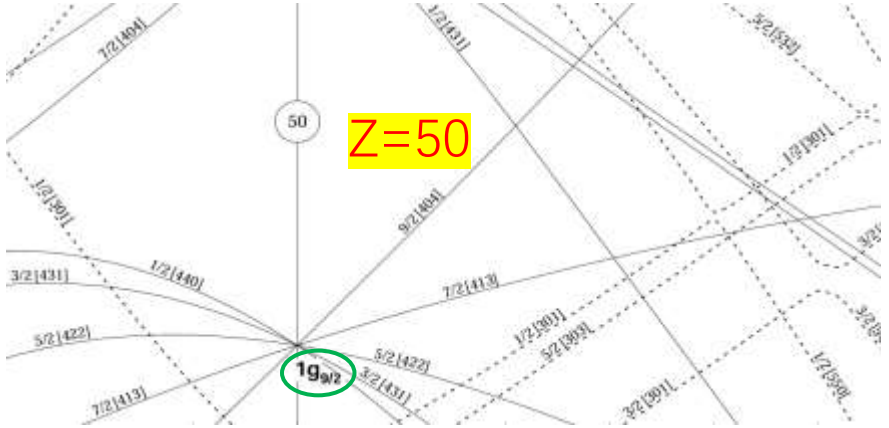
$$E'_{LN} = \sum_{j=1}^{S} e_{k_j} + 2\sum_{k\neq k_j} v_k^2 e_k - \frac{\Delta^2}{G} - G\sum_{k\neq k_j} v_k^4 + G\frac{N-S}{2} - 4\lambda_2 \sum_{k\neq k_j} (u_k v_k)^2$$

- k_j 标记被堵塞的轨道，S 为被堵塞的轨道数，N 是质子或中子数。
- 通过平均能隙法固定对力强度 G，对每个同位素分别确定

组态限制的势能面(PES)

- 核心特征是在最小化过程中固定单粒子组态。
- 在(β_2 , γ)网格上计算能量，并且在每个点上最小化 β_4 。
- 对于奇 A 核，未配对的质子会占据一个选定的轨道，在整个(β_2 , γ)平面上追踪该轨道。

$\Omega\pi\,[Nn_zm_l]$ asymptotic quantum numbers



- 轴对称形状下具有明确的总自旋投影 Ω；
- 但三轴形状 Ω 不再守恒，具有相似 |Ω| 的轨道可能接近或交叉。
- 为了追踪特定轨道，需要比较相邻网格点之间 Nilsson 量子数的期望值。

参数选择

- 采用形变且轴对称破缺的 Woods-Saxon 势，为这两种修正提供输入的单粒子能级，计算主要用 universal 参数。

	V_0 (MeV)	κ	a (fm)	r_0 (fm)	λ	$r_{0,so}$ (fm)
Neutron	49.6	0.86	0.70	1.347	35.0	1.31
Proton	49.6	0.86	0.70	1.275	36.0	1.32

- 该参数组主要拟合了 208Pb，以及 204Pb 和 212Rn 的高自旋谱，并能在广泛的质量范围内再现单粒子能级顺序和形变。

105Tc	4	highk1	0.33/0.19	-0.07
1/2+[420]	13			1
1/2+[431]	11			2
1/2+[440]	8			3
3/2+[431]	9			1 2
5/2+[422]	10			1 2

105Tc	103		0.33/-0.19	-0.05
5/2+[413]	10			02
7/2+[413]	9			02

105Tc	103	highk03	0.38/0.22	-0.05
1/2+[431]	11			02
3/2-[301]	11	11	12	01

105Tc	7	highk71
3/2+[431]	9	
5/2+[422]	10	
7/2+[413]	12	
1/2-[301]	13	
5/2-[303]	12	

105Tc	3		0.33/0.19	-0.05
3/2+[431]	9			31
5/2+[422]	10			31
7/2+[413]	12			31
1/2-[301]	13			31
5/2-[303]	12			31

数值计算

寻找轨道

```
*        SINGLE PARTICLE NUCLEAR SPECTRUM - SHELL MODEL
*
*
*
*      NILSSON NUMBERS FOR WS-BASIS


*     1)   -.3640E+02  + (  1)    0.3,  0.0,  0.0,  0.50
*     2)   -.3200E+02  - (  1)    1.3,  0.0,  1.0,  1.50
*     3)   -.3165E+02  - (  2)    1.2,  0.1,  1.0,  0.50
*     4)   -.2783E+02  - (  3)    1.2,  1.0,  0.0,  0.50
*     5)   -.2656E+02  + (  2)    2.2,  0.0,  2.0,  2.50
*     6)   -.2563E+02  + (  3)    2.2,  0.2,  1.9,  1.50
*     7)   -.2453E+02  + (  4)    2.1,  0.3,  0.2,  0.50
*     8)   -.2269E+02  + (  5)    2.2,  0.9,  1.1,  1.50
*     9)   -.2226E+02  + (  6)    2.1,  0.8,  0.8,  0.50
*    10)   -.2036E+02  - (  4)    3.2,  0.0,  3.0,  3.50
*    11)   -.1898E+02  - (  5)    3.1,  0.5,  2.6,  2.50
*    12)   -.1778E+02  - (  6)    3.1,  0.6,  1.4,  1.50
*    13)   -.1710E+02  - (  7)    3.0,  0.6,  0.6,  0.50
*    14)   -.1675E+02  + (  7)    2.2,  1.9,  0.0,  0.50
*    15)   -.1592E+02  - (  8)    3.1,  0.6,  2.4,  2.50
*    16)   -.1518E+02  - (  9)    3.0,  0.6,  1.5,  1.50
*    17)   -.1392E+02  - ( 10)    3.0,  0.7,  0.5,  0.50
*    18)   -.1358E+02  + (  8)    4.1,  0.0,  4.0,  4.50
*    19)   -.1207E+02  + (  9)    4.1,  0.7,  3.3,  3.50
*    20)   -.1115E+02  - ( 11)    3.1,  1.8,  1.2,  1.50
*    21)   -.1090E+02  - ( 12)    3.1,  1.9,  0.8,  0.50
*    22)   -.1079E+02  + ( 10)    4.1,  0.9,  2.4,  2.50
*    23)   -.9771E+01  + ( 11)    4.0,  1.0,  1.5,  1.50
*    24)   -.9164E+01  + ( 12)    4.0,  0.9,  0.5,  0.50
*    25)   -.7737E+01  + ( 13)    4.0,  0.3,  3.7,  3.50
*    26)   -.7633E+01  + ( 14)    4.0,  0.6,  2.3,  2.50
```

修改范围　修改起始点　修改参数

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echo
echo
highk                                  >>$FN_HIGHK_DATA
 \$DEFFI  NB2=10,  NGA=10,  BET20=0.26,  GAM0=0.10   NAZWIT=4,
          DB2=-0.02,  DGA=-0.02,  NNNSTP=2,  NNPSTP=2,
          DB4=$db4,  NNNDP=$DZ,  NNNGP=$GZ,  NNNDN=$DN,  NNNGN=$GN \$
 \$INPUT NMAXP=12,  NMAXN=14,  IZ0=$Z0,  IN0=$N0,  IDIM=135,
         OMCRN=0.22,  OMCRP=0.22,  OMEGIN=0.0,  IDECOM=0,
         STEPOM=0.05,  ILOM=1,  IPRINT=0,  NCUTDN=$CUTN1,  IEX=4,
         NCUTGN=$CUTN2,  NCUTDP=$CUTZ1,  NCUTGP=$CUTZ2,  FIXP=1.0,  FIXN=1.0,
         GFAC=1.0,  ITAUX=$ITAUX,  IPLOT=0,  ICHECK=0,  KPAIR=1,
         DFACTN=1.0,  DFACTP=1.0 \$
 \$SWITCH  ILIPEK=1,  IHART=1,  IDEFC=0,  B2CON=0.30,  GACON=24.0,
         B4CON=0,  G20F=0.0,  G21F=0.0,  G22F=0.0   \$
 \$CONFIG
          NQNPPS(1,1)=0,NQNPPS(1,2)=0,NQNPPS(1,3)=0,
          NQNPNS(1,1)=0,NQNPNS(1,2)=0,NQNPNS(1,3)=0,
          NQPPPS(1,1)=0,NQPPPS(1,2)=0,NQPPPS(1,3)=0,
          NQPPNS(1,1)=0,NQPPNS(1,2)=0,NQPPNS(1,3)=0,
          NQNPPS(2,1)=13,NQNPPS(2,2)=0,NQNPPS(2,3)=0,
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END_OF_INPUT
mv fort.10 $FN_HIGHK_DATA-$ind.dat
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填入

➢等待计算约 3 ~ 8min

数据处理

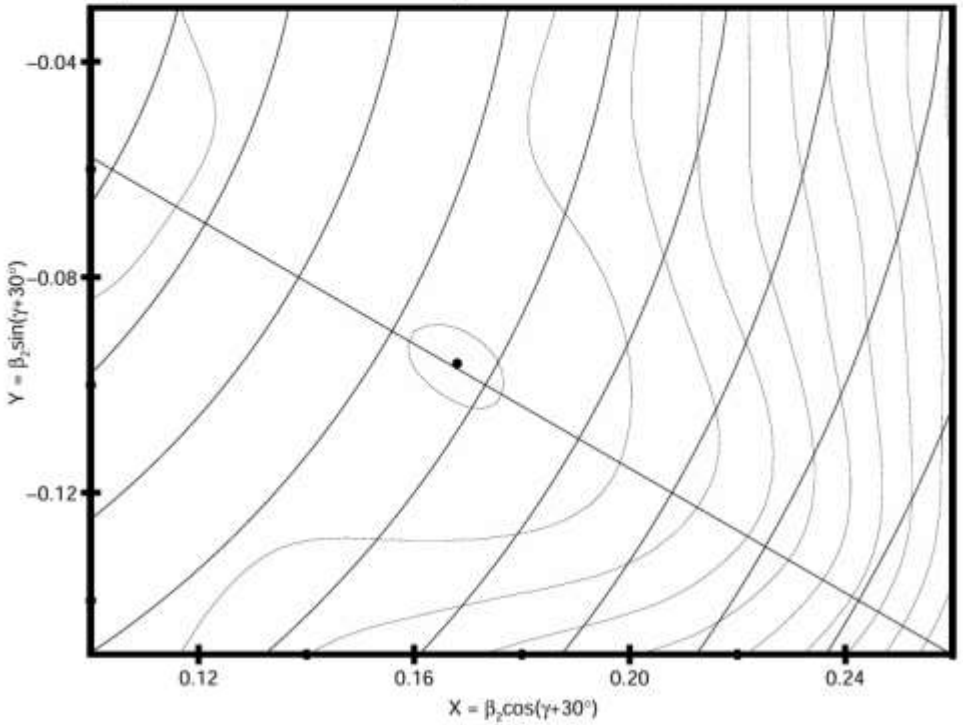
^{111}Ag ../sur/sur–z47–n64–111Ag.1.5+.

Z=47 N= 64 A=111 n: vacuum p: vacuum Δ(P+P*)

ω= 0.000 I= 0.000 Ip= 0.000 In= 0.000 E= 0.594

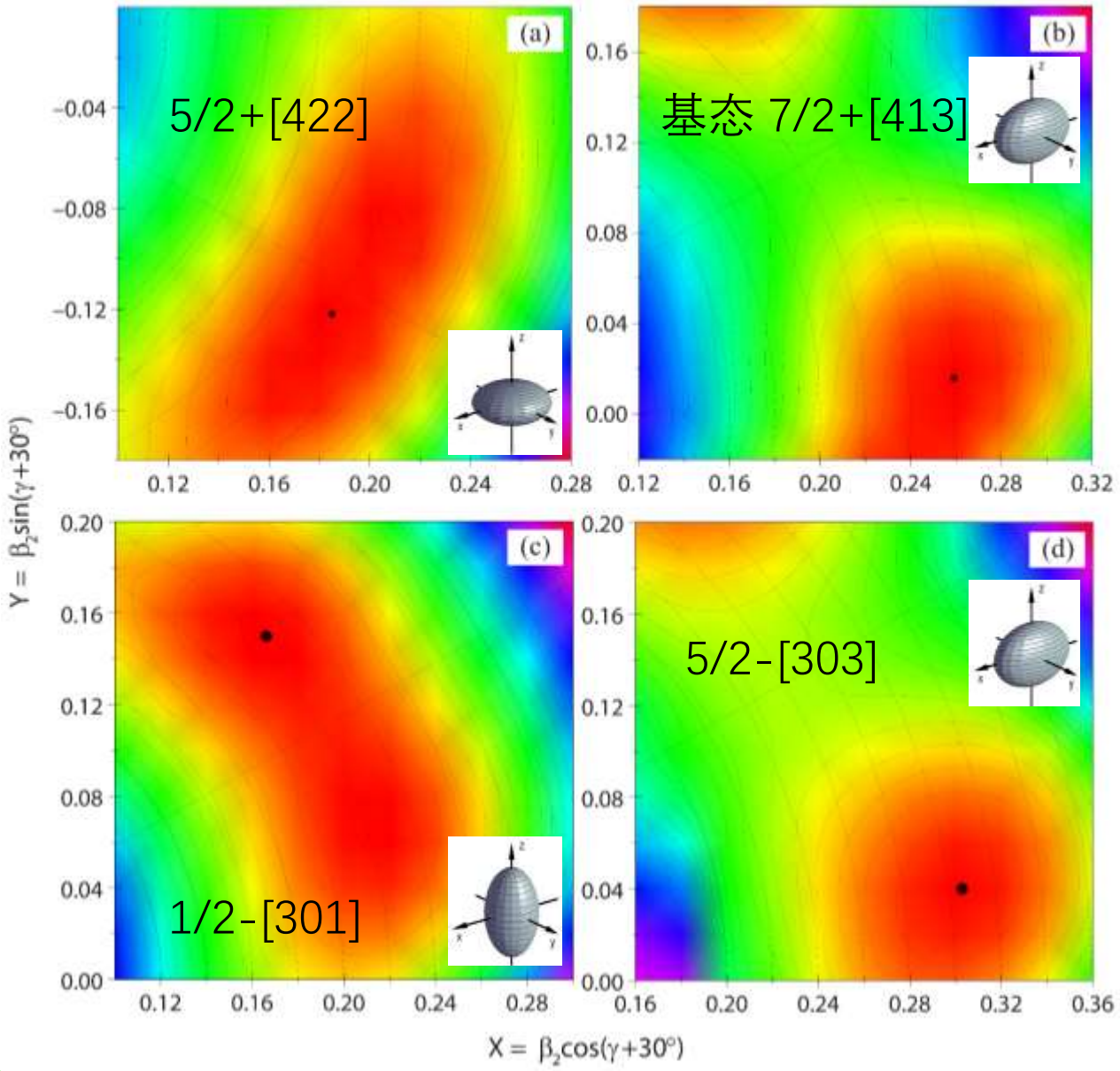
β_2= 0.193 γ= –59.724 β_4=–0.004 x= 0.168 y=–0.096

Δ_p=0.804 Δ_n=1.125 Δ_{psc}=1.125 Δ_{nsc}=0.804



Nuclei	K^π	Configuration	E_{exp}(keV)	E_{cal}(keV)	β_2	β_4	γ
^{109}Ag	$7/2^+$	$\pi 7/2^+[413]$	88	0	0.159	−0.003	0
	$1/2^-$	$\pi 1/2^-[301]$	0	265	0.188	−0.011	−20
	$1/2^+$	$\pi 1/2^+[431]$		820	0.248	0.031	1
	$3/2^+$	$\pi 3/2^+[431]$	724	1023	0.065	0.001	20
^{111}Ag	$7/2^+$	$\pi 7/2^+[413]$	60	0	0.166	−0.005	0
	$1/2^-$	$\pi 1/2^-[301]$	0	182	0.198	−0.012	−20
	$3/2^+$	$\pi 3/2^+[431]$	377	448	0.193	0	−60
	$1/2^+$	$\pi 1/2^+[431]$		642	0.276	0.006	−18
^{113}Ag	$7/2^+$	$\pi 7/2^+[413]$	44	0	0.170	−0.008	0
	$1/2^-$	$\pi 1/2^-[301]$	0	124	0.184	−0.009	−40
	$3/2^+$	$\pi 3/2^+[431]$	222	134	0.208	−0.015	−60
	$5/2^+$	$\pi 5/2^+[422]$	477	606	0.154	−0.001	−61
	$1/2^+$	$\pi 1/2^+[431]$		606	0.281	−0.005	−20
	$7/2^+$	$\pi 7/2^+[413]$	370	853	0.059	−0.005	−76

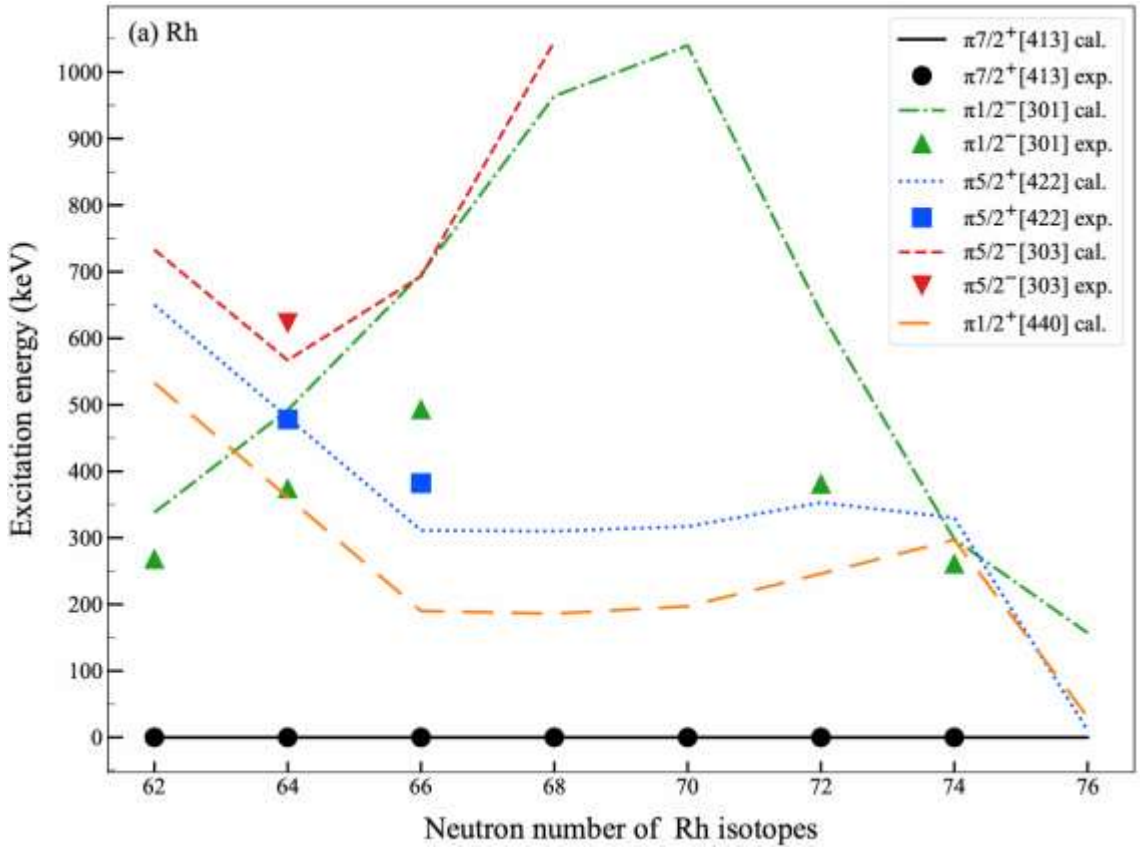
合成彩图

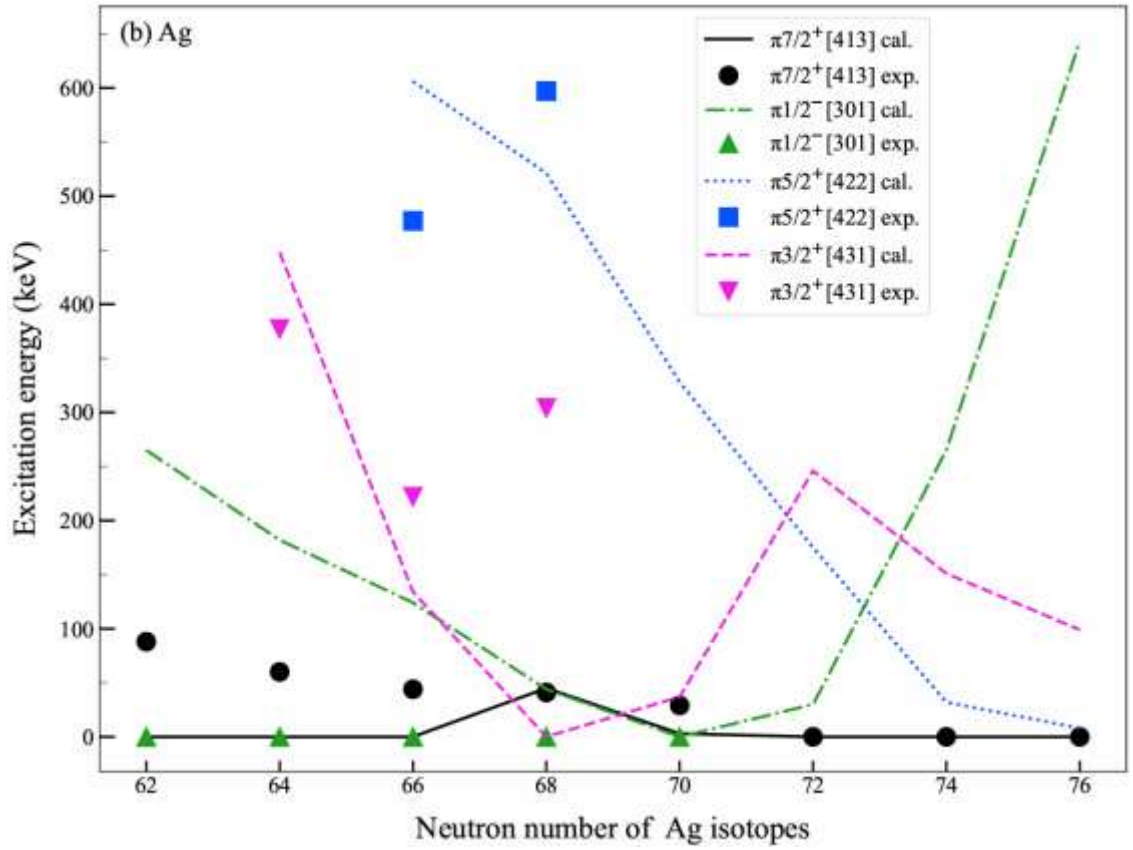


- 109Rh 中的形状共存

- 基态 7/2+[413]， 三轴形变
- 扁椭 5/2+[422] : E = 481 keV
- 长椭 1/2-[301] : E = 492 keV
- 三轴 5/2-[303] : E = 567 keV

能量曲线





- Rh 和 Ag 的同位素链中，部分组态的能量与实验相比较

实验合作

① 190At

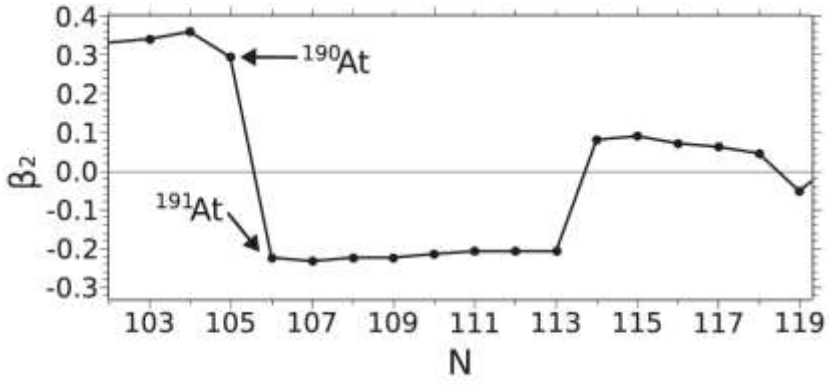
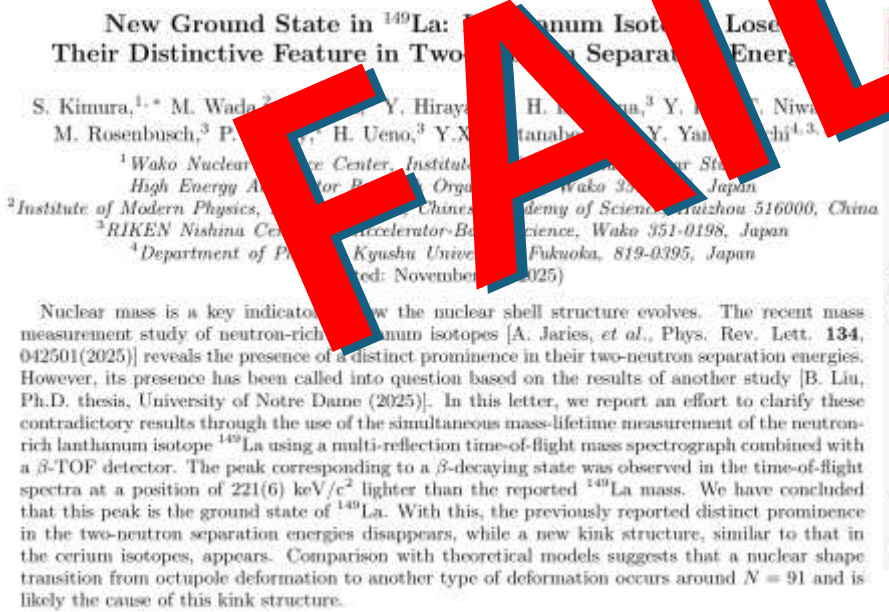


FIG. 1. Theoretical quadrupole ground-state deformation β_2 values for $^{188-204}$At isotopes based on the macroscopic-microscopic model [8].

② 149La



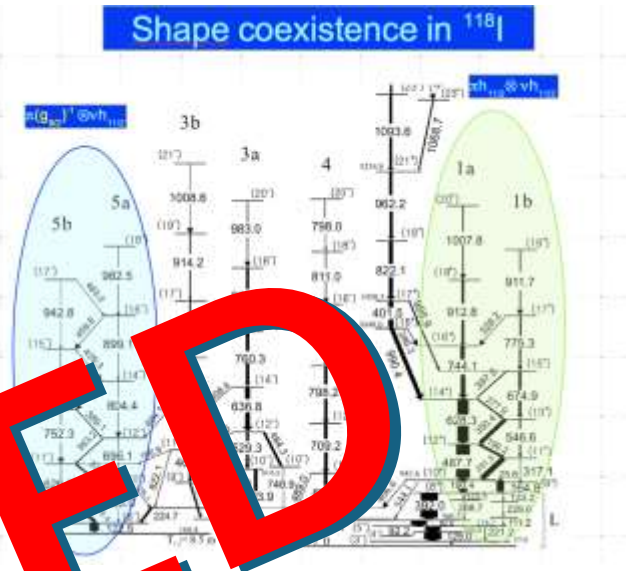
New Ground State in ^{149}La: ... num Isot... Lose... Their Distinctive Feature in Two... Separa... Energ...

S. Kimura,[1,*] M. Wada,[2] ... Y. Hiraya... H. ... ua,[3] Y. ... Niwa... M. Rosenbusch,[3] P. ..., H. Ueno,[3] Y.X. ...tanaby... Y. Yan... hi[4,3]

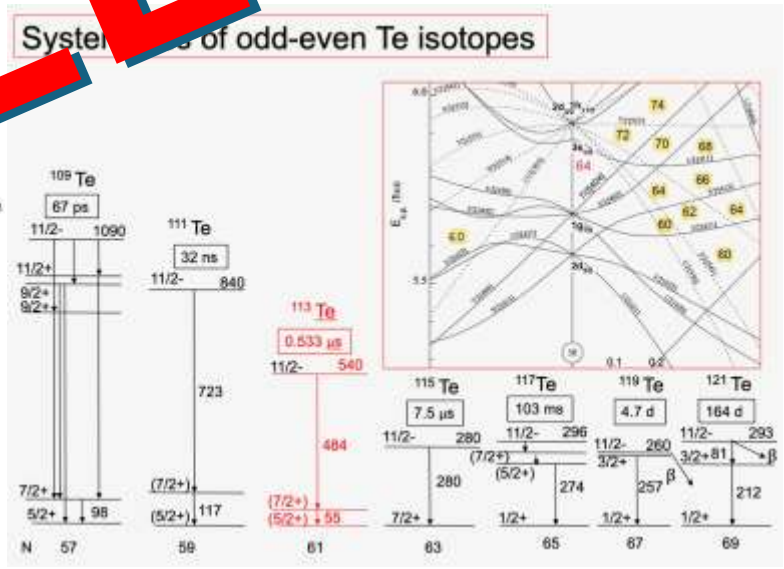
[1] *Wako Nuclear ... Center, Institute ... Wako 3... Japan*
[2] *Institute of Modern Physics, ... Chines... demy of Science, Huizhou 516000, China*
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[4] *Department of P... Kyushu Unive... Fukuoka, 819-0395, Japan*
...ed: November ...025)

Nuclear mass is a key indicato... w the nuclear shell structure evolves. The recent mass measurement study of neutron-rich ...num isotopes [A. Jaries, *et al.*, Phys. Rev. Lett. **134**, 042501(2025)] reveals the presence of a distinct prominence in their two-neutron separation energies. However, its presence has been called into question based on the results of another study [B. Liu, Ph.D. thesis, University of Notre Dame (2025)]. In this letter, we report an effort to clarify these contradictory results through the use of the simultaneous mass-lifetime measurement of the neutron-rich lanthanum isotope ^{149}La using a multi-reflection time-of-flight mass spectrograph combined with a β-TOF detector. The peak corresponding to a β-decaying state was observed in the time-of-flight spectra at a position of 221(6) keV/c^2 lighter than the reported ^{149}La mass. We have concluded that this peak is the ground state of ^{149}La. With this, the previously reported distinct prominence in the two-neutron separation energies disappears, while a new kink structure, similar to that in the cerium isotopes, appears. Comparison with theoretical models suggests that a nuclear shape transition from octupole deformation to another type of deformation occurs around $N = 91$ and is likely the cause of this kink structure.

③ 118I



④ 109~121Te



FAILED

研究结论

- 利用组态限制的势能面计算，研究了奇 A 的107 ~ 127Rh 和 109 ~ 133Ag 中的低能单准质子能级带首。

- 经系统学分析，与 n7/2+[413]、n1/2−[301]、n5/2+[422]、n3/2+[431] 态的现有数据的主要趋势相符。两个同位素链中都发现了强烈的组态依赖形状极化。

- 随着中子数增加，Rh 中 N = 76 和 Ag 中 N = 74 处，最低的正宇称极小值变得接近球形，而 n1/2−[301] 组态仍然保持明显的形变。